COMSATS UNIVERSITY,ISLAMABAD

Department of Computer Science
Theory Assignment-2,Spring-2026
Submission Date: 16/10/2026

Course: Database Systems (CSC270) **Class:------BSCS-III**

Instructor:-Dr. Rubina Adnan **Total Marks:10**

Mapped to CLO2 [Apply data modeling and normalization techniques to design database for small to medium size enterprise]

For each of the following problems, first create an entity relationship (ER) diagram and then convert it into a relational database schema. Assume attributes as per your real-world knowledge for the given scenarios,if not mentioned.

Problem-1: Bioinformatics Database [1]

Draw an ER diagram for the following Bioinformatics application.

Patient: has a unique MSP number, a Patient name, a Date of Birth, a Tissue Type and an indicator denoting whether the tissue is cancerous or normal. A patient library associates a patient with multiple tags. Each tag has a unique tag number and a unique nucleotide sequence.
For each tag in the patient library, a count is given to record the number of times the tag occurs in the library. In general, the same tag can be associated with any number of patients. A tag may be mapped to a gene. Each gene has a unique gene name and a type. In general, multiple tags may be mapped to the same gene. However, two different Genes cannot be mapped to the same tag.

Finally, an article is identified by a unique article number and a journal name. An article may analyze multiple genes and a gene may be analyzed by multiple articles.

Problem-2: Blue Wave Water Sports Center Database [2]

Blue Wave Water Sports Center operates at a beach resort and provides two main services to tourists-Jet Ski rentals and Surfboard sales. The management wants to automate their business process for handling both rentals and sales using a database system. The following requirements have been gathered:

1.Equipment. The center offers two types of equipment: Jet Skis (for rent) Surfboards (for sale). Each piece equipment is identified by a unique Equipment ID. For both equipment types, their Model Name must be stored. For Jet Skis,the system must store the rental price per hour. For Surfboards, the system must store the sale price.

2. Tourists. Tourists are identified by their Passport Number. The system must also record their: Full Name, Nationality And Multiple Contact Numbers.

3.Tourist Types: There are two types of tourists: Riders→ those who rent Jet Skis and Surfers →those who buy Surfboards. For Riders, the system should also record their riding experience (in years).

4. Rentals: A Rider can rent one or more Jet Skis. Each Jet Ski can be rented by many Riders over time. For every rental, the system must record: Rental Date and Return Date.

5.Sales: A Surfer can purchase one or more Surfboards. Each Surfboard can be purchased by only one Surfer (once sold). For every purchase, the system must record:Purchase Date.

Problem-3: Football Match Database [3]

The database for Football Match needs to be built. The following set of requirements have been identified:

- Football Clubs play Games. Each football club have specific id, name, description, color, and other details.
  Football Clubs have specific Uniforms. Each Uniform has unique identification number, name, and description.

- Football Clubs have specific Jargon, such a 'Fumble' in the NFL. Jargon have id, phases and meaning.
  - Football Clubs have a range of Reference Material, such as Footballer Contracts. Reference identification,reference material title and description.
  - Games always involve two Football Clubs. Games have different details regarding game like date,results,id,and description.
- Games are played at different Locations. Each location has id, name, and address.
- Games always produce Deliverables-such as Goals and Results.Deliverables has name, descriptions, and id.
  - Football Clubs have Cub Personnel, includng Players. Club personals have name, title, gender, address,email address,contact details etc.
- Players are involved in Games as Game Players.
  Players in Games can play at specific Positions. Each position has certain details.
  - Games can involve Value Systems, such as 'Play by the Rules and Win'. Value system has id, name and description.

Problem-4: Chughtai Group of Pharmacies Database [4]

You are to design a normalized relational database for Chughtai group of pharmacies. Here are the requirements provided to you:

Patients are identified by their CNICs. For each patient, the name, address, and age must be recorded.

Doctors are identified by their CNICs. For each doctor, the name, specialty, and years of experience must be recorded.

- Every patient has a primary doctor. Every doctor has at least one patient.
  Each pharmaceutical company is uniquely identified by name and has a phone number.
  - For each drug, the trade name and formula must be recorded. Each drug is sold by a given pharmaceutical company,and the trade name identifies a drug uniquely from among the products of that company only and not across all the companies.
- Each pharmacy has a unique name, address, city, and phone number.
  Each pharmacy sells several drugs and has a price for each. A drug could be sold at several pharmacies, and the price could vary from one pharmacy to another.
  - Doctors prescribe drugs for patients. A doctor could prescribe one or more drugs for several patients, and a patient could obtain prescriptions from several doctors. Each prescription is assigned an ID, has a date and,a quantity associated with it.

Problem-5: University Database [5]

Construct an ER diagram for the givenscenario: A university has several departments. Each department is managed by a Faculty.Each faculty is assigned to only one department. Faculty are of different designations as given

Assistant Professor,Associate Professor and Professor. Only Assistant Professors are assigned courses.Every

Assistant professor can teach maximum of five courses. A course is taught by a maximum of one assistant Professor.

Only Associate Professors can conduct seminars. Professors can only work on funded projects.Only faculty who is a professor can manage a department